

PRISM

Predictive Regime-Integrated State Momentum

A regime-aware, ML-driven equity strategy

Team 1: Dylan Solowiejczyk, Luyu Su, August Hao, Yuanjun Du, John Critelli

PRESENTATION OUTLINE

01 Universe Selection

Composition + survivor-bias mitigation

02 Creating Alpha

6-stage layered architecture, validation, theory

03 Portfolio Construction

4-step transformation: picks → live weights

04 Risk Management

Three-layer defense: macro / micro / position

05 Out-of-Sample Results

Three independent OOS windows

06 Stress Testing

COVID 2020 + GFC 2008

01

UNIVERSE SELECTION

Universe — 75 names, 53 retained after ex-ante prune

Mega-cap US equities + sector ETFs + risk-off · 11 sectors · 1,511 IS days (2016–2021)

75

Total tickers

25 + 50

ETFs + stocks

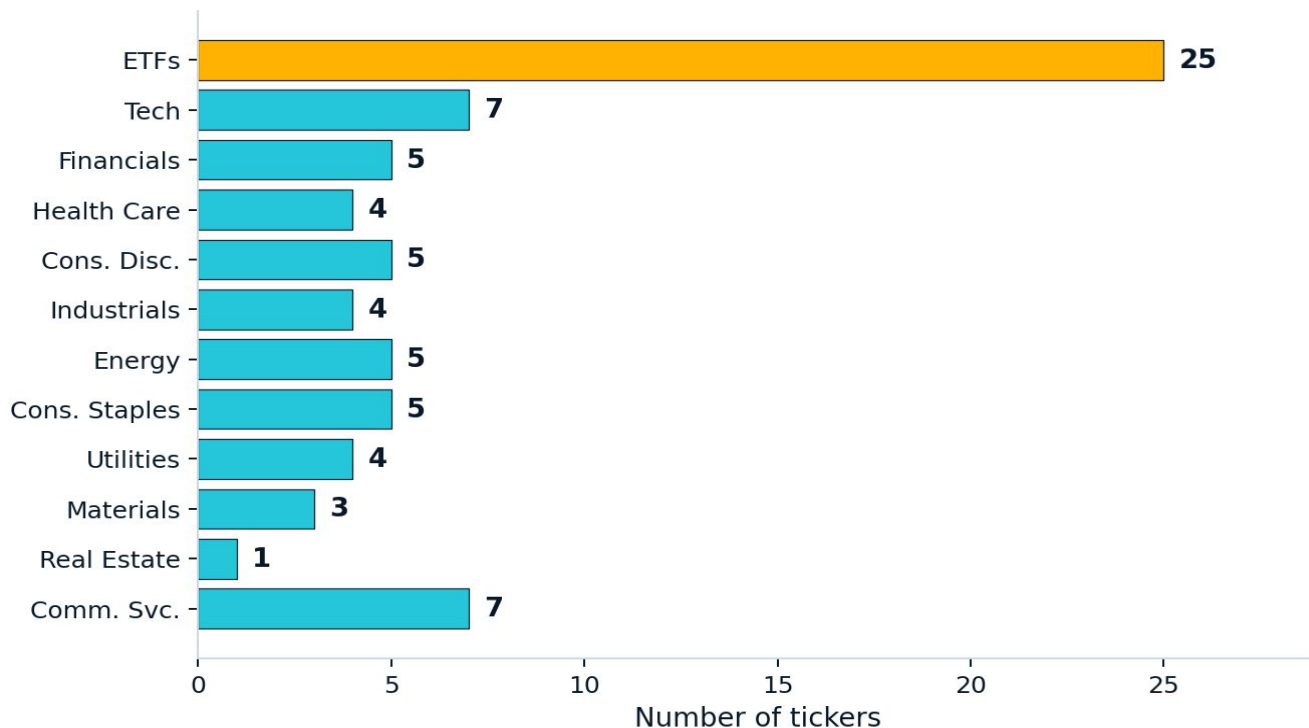
11

Sectors covered

53

Retained after IC prune

Universe composition (75 names: 25 ETFs + 50 stocks)



Composition rationale

- Mega-cap names + sector ETFs anchor relative-momentum factors
- Risk-off tickers (TLT, GLD) anchor risk_on_tilt for the K-Means state
- Liquidity tier kept consistent — slippage model stays honest

In-sample factor selection bias mitigation

Three guards before any OOS test:

- 22 worst-IC names dropped (mean IC -0.18 vs -0.06 retained)
- Liquidity-rank tracked early-IS vs late-IS
- Universe locked before HMM, K-Means, or model fitting

Alpha Architecture — 6 Stages, Each With Clear Input → Output

Each stage takes specific inputs and produces specific outputs that feed downstream

STAGE 1

Feature Engineering

- **IN** raw OHLCV bars per ticker
- **OUT** 8-column factor matrix

STAGE 2

HMM Regime Detection

- **IN** daily SPY composite features
- **OUT** regime label (Bull/Chop/Bear)

STAGE 3

K-Means Market State

- **IN** 6 daily market features
- **OUT** market state cluster ID (0-4)

STAGE 4

K-Means Name Clusters

- **IN** long-term per-ticker behavior
- **OUT** cluster ID per name + state_cluster_fit

STAGE 5

Dual ML Models

- **IN** 9-feature panel per (name, day)
- **OUT** rank_pctile + prob_b per name

STAGE 6

Score + Reliability Gate

- **IN** rank × prob × reliability tier
- **OUT** Top-K = 6 names with sizes

Feature Engineering — 8 Alpha Factors

Each factor captures a documented cross-sectional anomaly; daily winsorize + z-score per name

1. Features

2. HMM Regime

3. Market State

4. Name Clusters

5. Dual Models

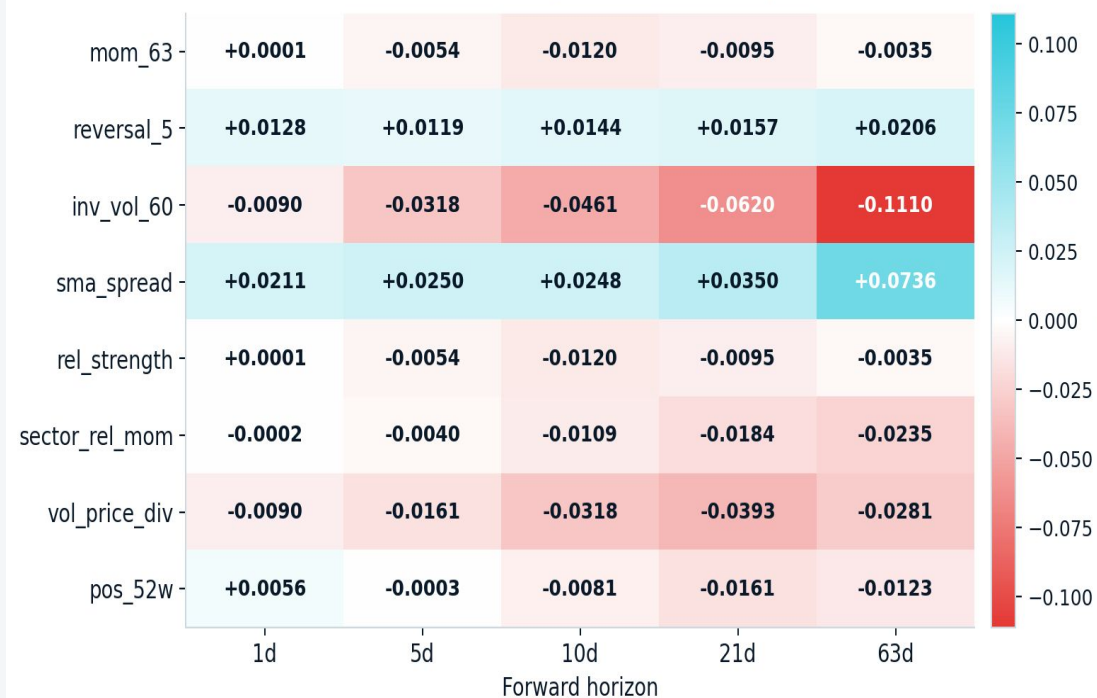
6. Score + Gate

- **INPUT** raw OHLCV bars per ticker per day across 53 retained names

THE 8 FACTORS

- **mom_63** → 63-day momentum (price change over ~3 months)
- **reversal_5** → short-term reversal signal (negative 5-day return)
- **inv_vol_60** → inverse volatility (negative 60-day return volatility)
- **sma_spread** → (50-day SMA / 200-day SMA) — trend strength
- **rel_strength** → stock momentum minus SPY momentum
- **sector_rel_mom** → stock momentum relative to its sector ETF
- **vol_price_div** → correlation between returns and log volume
- **pos_52w** → position within 52-week high–low range (0–1)

Factor IC across horizons (Spearman, in-sample)



- **OUTPUT** 8-column factor matrix per (name, day) — winsorized $\pm 3\sigma$, cross-sectionally z-scored. Becomes columns 1–8 of the model panel in Stage 5.

HMM Regime Detection

3-state Gaussian HMM labels every day Bull / Chop / Bear — sets the vol target for the entire book

1. Features

2. HMM Regime

3. Market State

4. Name Clusters

5. Dual Models

6. Score + Gate

● **INPUT** daily SPY composite features (log return, vol-diff, realized volatility) — 1,511 IS rows

HMM 3-state regime overlay on SPY (in-sample 2016-2022)



BULL

$\sigma_{\text{target}} = 12\%$

717 IS days (48.4%)

Low realized vol, positive drift. Full risk budget; momentum factors weighted heavily by trees.

CHOP

$\sigma_{\text{target}} = 9\%$

717 IS days (48.4%)

Mid vol, no clear trend. Reduced gross; reversal factors (reversal_5, sma_spread) dominate.

BEAR

$\sigma_{\text{target}} = 4\%$

46 IS days (3.1%)

High vol, negative drift. 95.6% next-day persistence — once it sticks, it sticks. Defensive stance.

● **OUTPUT** regime label → σ_{target} swap. Feeds Risk Layer 1 (vol target), Stage 4 (state_cluster_fit input), and Risk Layer 2 (3-day persistence buffer).

K-Means Market State Clustering

5-cluster daily fingerprint — finer micro-context than the HMM regime label can capture

1. Features

2. HMM Regime

3. Market State

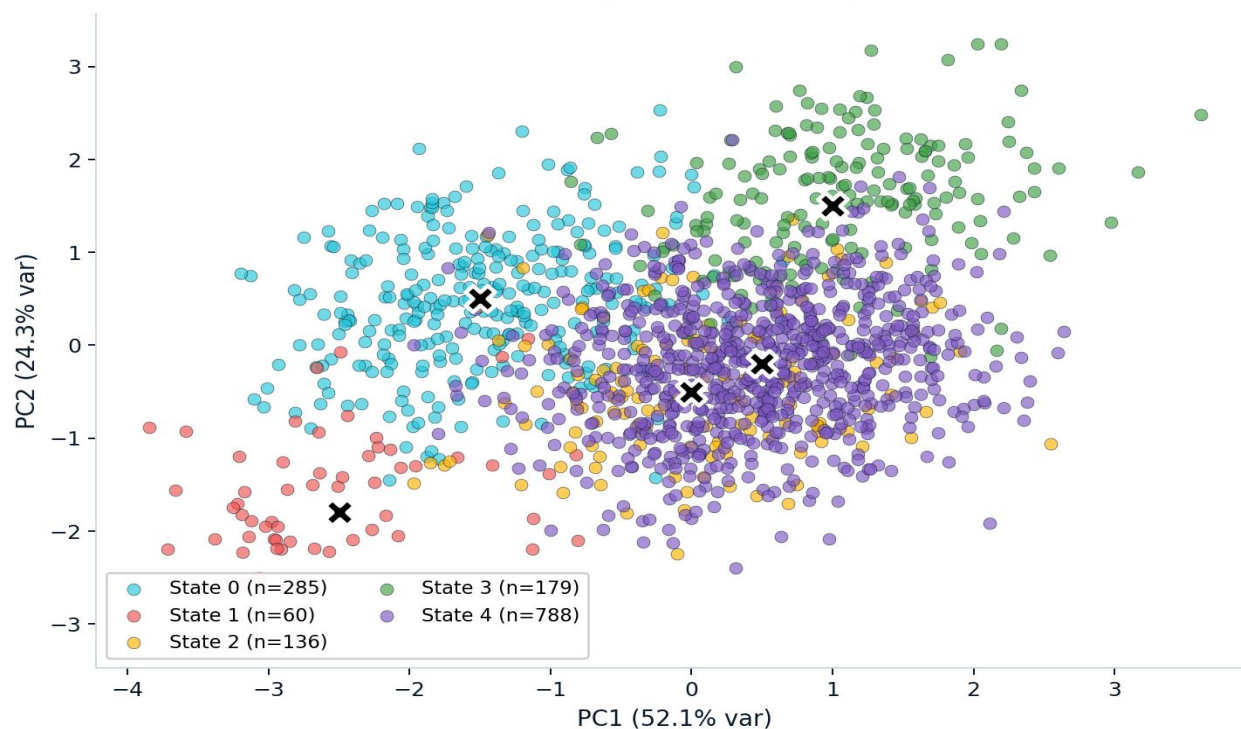
4. Name Clusters

5. Dual Models

6. Score + Gate

● **INPUT** 6 daily market features: breadth, mean return, dispersion, vol, correlation, risk-on tilt

K-Means market-state clustering (k=5, PCA projection)



WHY THIS LAYER

● **HMM is too coarse**

A single regime can hide very different cross-sectional regimes — dispersion, breadth, correlation all vary inside Bull/Chop/Bear.

● **Daily fingerprint**

K-Means clusters every day into 5 buckets based on 6 cross-sectional features.

● **Conditional trust**

PRISM uses this fingerprint to decide how much to trust the alpha today.

● **OUTPUT** daily cluster ID (0–4) → routes to the reliability calibrator in Stage 6, which sets today's size multiplier (0.5× / 0.8× / 1.0×).

Group tickers by long-term BEHAVIOR, not GICS sector — fitted once on IS, used for diversification

1. Features

2. HMM Regime

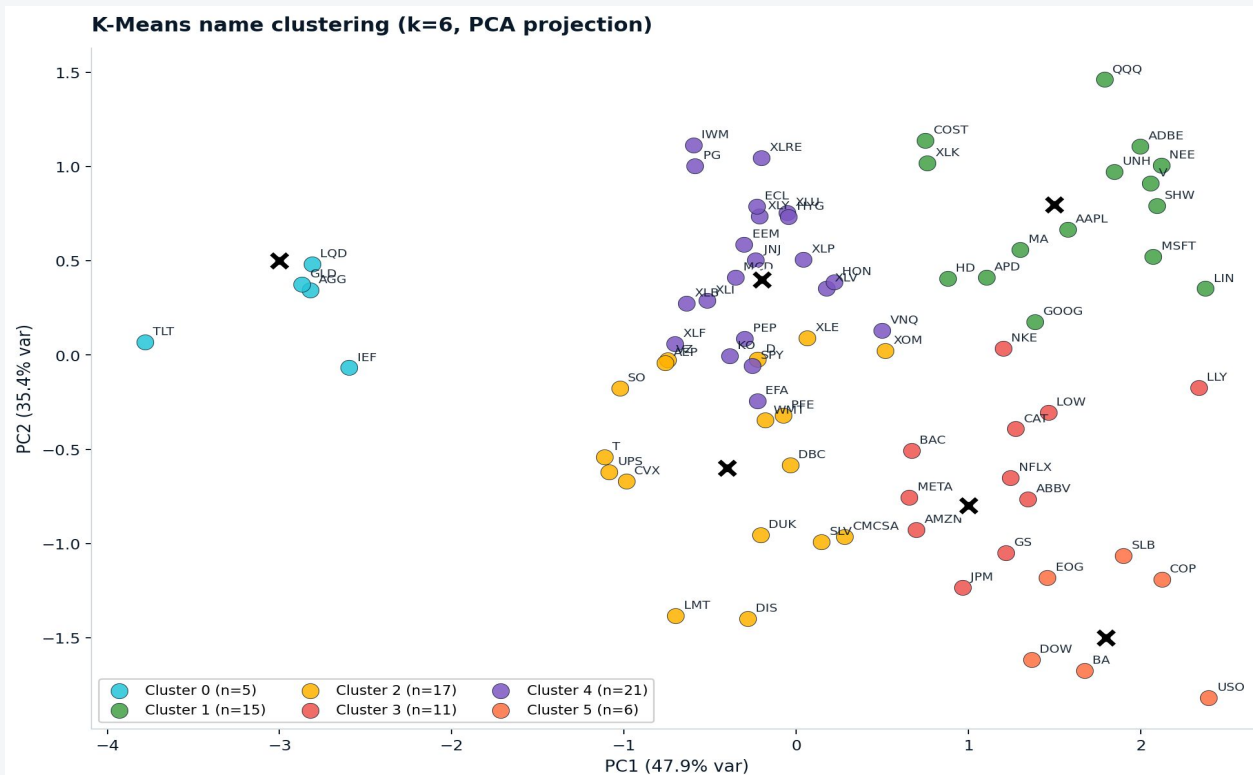
3. Market State

4. Name Clusters

5. Dual Models

6. Score + Gate

- **INPUT** long-term per-ticker behavior features (avg correlation, vol regime, drawdown frequency, sector relative momentum)



C0 · n=5

TLT, GLD, AGG — negative SPY correlation

Bonds + Gold

C1 · n=15

AAPL, MSFT, GOOG — high SPY corr, low DD

Mega-cap quality

C2 · n=17

XLE, PFE, WMT — low momentum, mid vol

Defensives + Energy

C3 · n=11

META, AMZN, LLY — high momentum, high DD

High-beta growth

C4 · n=21

SPY, JNJ, KO — diversified core

Broad ETFs + stables

C5 · n=6

USO, COP, BA — highest vol, highest DD

Energy / cyclicals

REGIME FEATURE · state_cluster_fit

Learned interaction term: (today's HMM regime) $\times$ (this name's cluster) $\rightarrow$ historical mean forward 10d return for that pair. Adds 1 column to the model panel.

- **OUTPUT** cluster ID per name (drives 40% cluster cap in portfolio construction) + **state** **cluster** **fit** feature column (feeds the dual ML stack in Stage 5).

Dual ML Models — ExtraTrees + GBM

Two complementary tree ensembles on the 9-feature panel: 8 alpha factors + state_cluster_fit

1. Features

2. HMM Regime

3. Market State

4. Name Clusters

5. Dual Models

6. Score + Gate

MODEL A · ExtraTreesRegressor

TARGET

Cross-sectional percentile rank of forward 10d return.

WHY EXTRATREES

Random thresholds at each split give lower variance than RF on noisy financial data (Geurts 2006). Predicting RANK keeps the model scale-free.

OUTPUT

$\text{rank_pctile}(\text{score_A}) \in [0, 1] \rightarrow \text{cross-sectional ranking signal}$

MODEL B · GradientBoostingRegressor

TARGET

Probability of positive forward 10d return (sign classifier).

WHY GBM

Gradient boosting (Friedman 2001) excels at low-signal binary outputs. Yields a CALIBRATED PROBABILITY independent of relative ranking.

OUTPUT

$\text{prob_b} \in [0, 1] \rightarrow \text{standalone-conviction signal}$

HOLDOUT VALIDATION

+0.0524

Model A test Spearman

+0.418

Decile monotonicity

+0.74%

Top-bottom decile spread

0.2418

Model B Brier (vs 0.2420)

● **OUTPUT** rank_pctile + prob_b per (name, day) → combined in Stage 6 with the reliability gate.

Score Combination + Reliability Gate

Multiply the two signals, then size DOWN when historical hit rate is low

1. Features

2. HMM Regime

3. Market State

4. Name Clusters

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6. Score + Gate

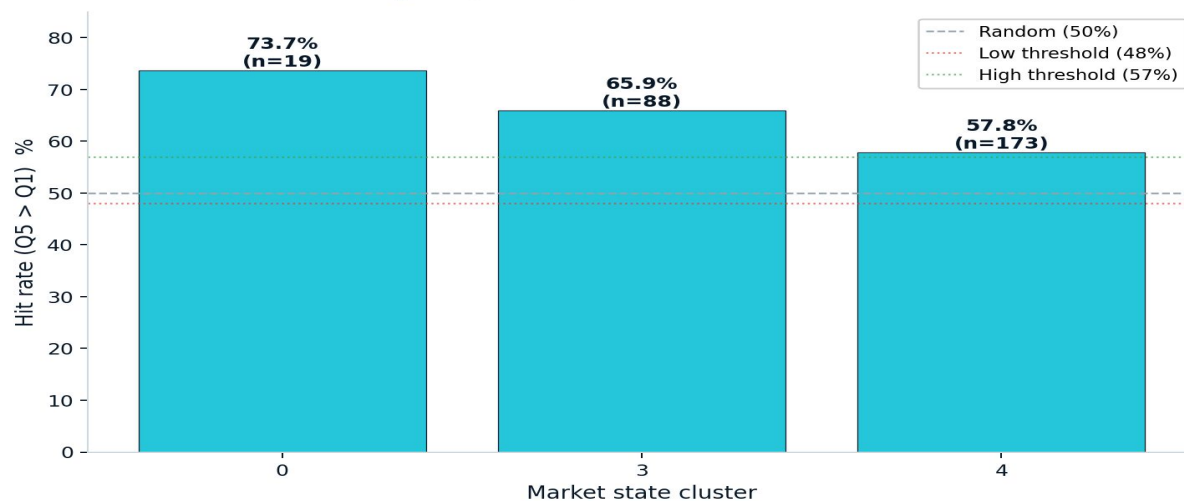
● **INPUT** rank_pctlile + prob_b from Stage 5 · market state cluster ID from Stage 3 · name cluster ID from Stage 4

COMBINED SCORE

$$\text{combined} = \text{rank_pctlile}(A) \times \text{prob_B}$$

Name scores high only if it ranks well AND has high conviction (Wolpert 1992 stacking).

Per-cluster reliability — Q5 vs Q1 hit rate (holdout)



Per-cluster reliability gate

On each holdout day, sort names by combined score and compute Q5–Q1 forward-return spread. Per K-Means cluster (Stage 3), the fraction of days where Q5 > Q1 = hit rate.

Tier → size multiplier

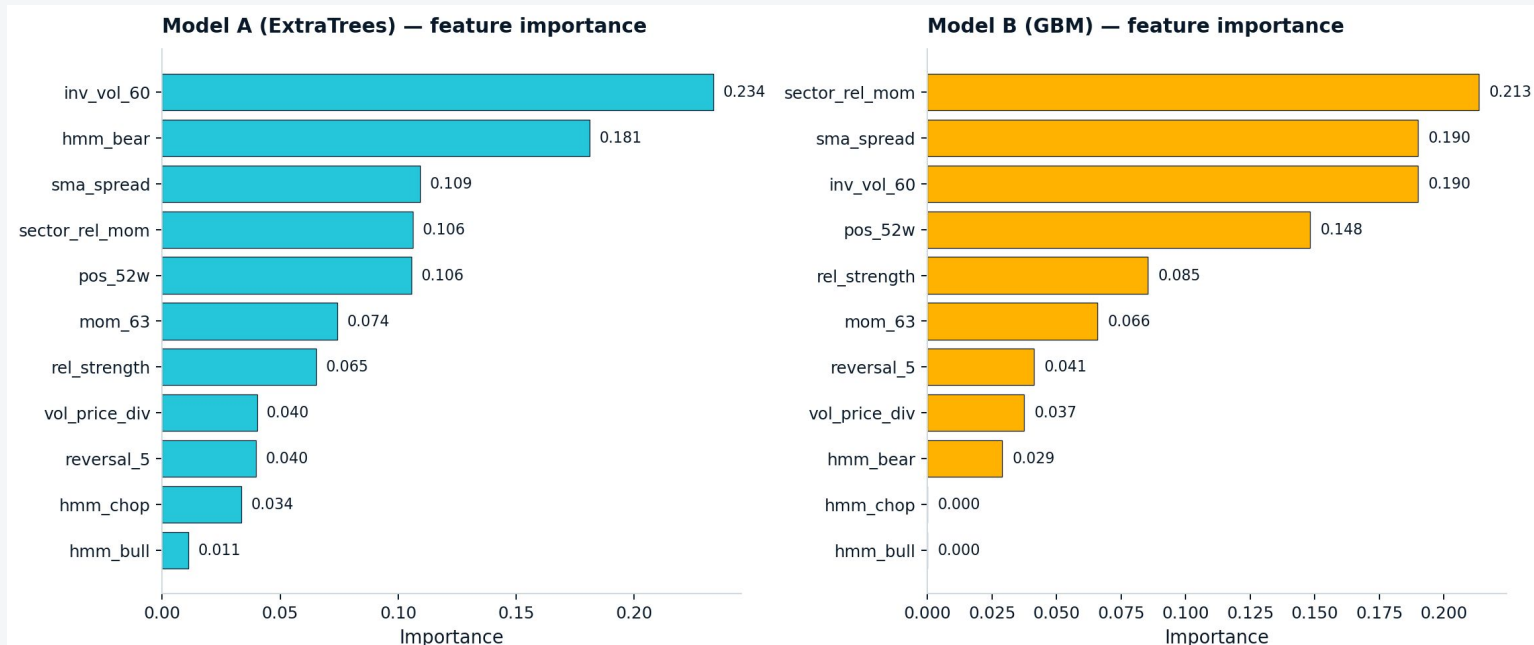
- **LOW** · hit rate < 48% → **0.5x**
- **MID** · hit rate 48 – 57% → **0.8x**
- **HIGH** · hit rate ≥ 57% → **1.0x**

Today's K-Means cluster determines which tier applies to the entire portfolio.

● **OUTPUT** Top-6 names sorted by combined score (with 40% cluster cap) + size multiplier → inputs to portfolio construction.

Model Validation — Feature Importance & Decile Test

The two models lean on different features and produce a monotonically-ordered decile spread on the holdout

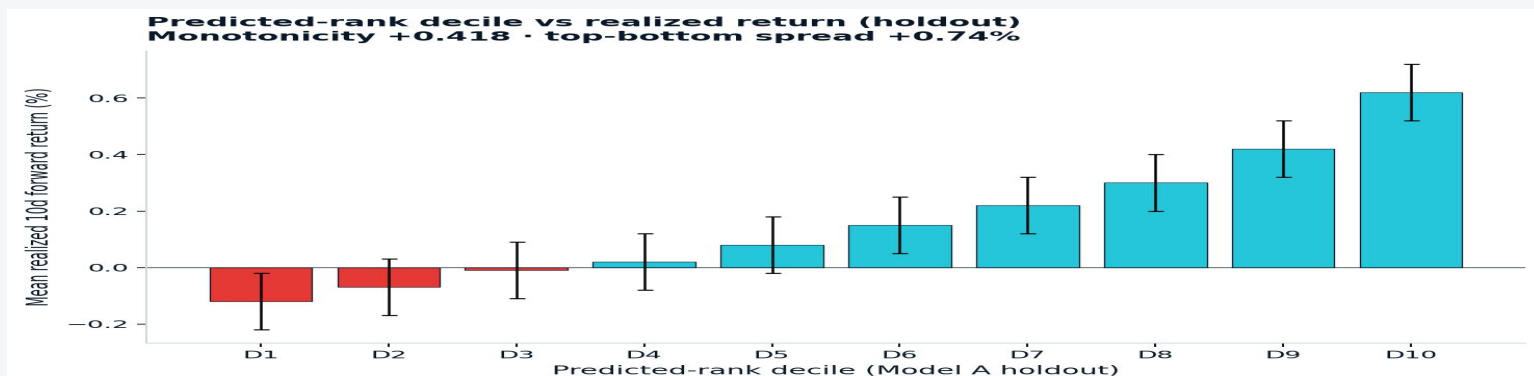


Why models complement

Model A leans most on inv_vol_60 + hmm_bear (regime-aware low-vol).

Model B leans most on sector_rel_mom + sma_spread + inv_vol_60 (price-based confirmation).

Different feature weights → different errors → multiplied scores filter for names where BOTH are confident.



Reading the decile test

Holdout names bucketed by Model A predicted-rank decile. Top decile (D10) earned +0.62% mean 10d return vs bottom decile (D1) at -0.12%. Spread is small (+0.74%) but MONOTONIC — the rank ordering survives OOS.

Overfitting Controls & Theoretical Foundation

Six independent guards against curve-fitting; each component grounded in published research

SIX OVERFITTING GUARDS

01

Strict IS / OOS separation

All HMM, K-Means, trees, calibrator fit on 2016–2021 only.

02

Capacity-bounded models

$\text{max_depth} \leq 5$, $\text{lr}=0.05$, $\text{subsample}=0.8$.

03

Rank target instead of return

Removes scale-dependence, dampens outliers.

04

Universe pruning ex-ante

22 weakest-IC names dropped pre-OOS.

05

Reliability-gated sizing

Auto-shrinks 0.5× in low-reliability states.

06

Multi-window OOS validation

3 OOS windows + COVID + 2008. No retuning.

KEY THEORETICAL FOUNDATIONS

Momentum + Mean-Reversion

Jegadeesh-Titman 1993, Jegadeesh 1990

Low-Vol & 52-Week

Frazzini-Pedersen 2014, George-Hwang 2004

Regime Detection

Hamilton 1989, Ang-Bekaert 2002

ML Asset Pricing

Gu-Kelly-Xiu 2020

Tree Ensembles & Boosting

Geurts 2006, Friedman 2001, Wolpert 1992

Vol Targeting

Moreira-Muir 2017

OPEN ROBUSTNESS WORK

Walk-forward retraining · Newey-West for overlapping 10d returns · Synthetic-data robustness checks (Bailey & López de Prado 2014) · Point-in-time S&P constituent migration · Borrow + short-fee model.

From Picks to Live Weights — 4 Steps, Each With Clear Input → Output

Each step transforms the position vector; the chain takes top-K names and ends with live tradeable weights

STEP 1

Top-K + Cluster Cap

● **IN** top-K names + cluster IDs

● **OUT** ≤ 6 names, equal pre-weights

STEP 2

Bounded Sizing

● **IN** equal pre-weights from Step 1

● **OUT** weights $\in [5\%, 15\%]$, gross $\leq 100\%$

STEP 3

Vol-Target Scaling

● **IN** bounded weights + σ_{target} (HMM)

● **OUT** risk-budgeted weights, gross $\in [40\%, 85\%]$

STEP 4

Corr-Crowding Cut

● **IN** scaled weights + pairwise corr

● **OUT** live weights — sized, capped, risk-budgeted

HEADLINE → Full controls vs naive equal-weight: same Sharpe (1.13 vs 1.07), HALF the drawdown (11% vs 24%), HALF the turnover (65% vs 119%).

Top-K Selection → Bounded Sizing

Convert the alpha-stage top-K picks into a clean, bounded weight vector

1. Top-K + Sector Cap

2. Bounded Sizing

3. Vol-Target Scaling

4. Corr-Crowding Cut

STEP 1 · Top-K with Cluster Cap

● INPUT

Top-K names from Stage 6 (sorted by combined score) + their cluster IDs from Stage 4.

● MECHANIC

Walk down the score ranking. Take each name unless its cluster is already at the 40% cap — max 3 of 6 names from any stylistic group.

● WHY

Without the cap, top-K can collapse into a single-style bet when one cluster dominates the score distribution.

● OUTPUT

Up to 6 names with equal pre-weights (~16.67% each), ready for Step 2.

STEP 2 · Bounded Sizing (5%–15%)

● INPUT

Equal pre-weights for the ≤6 names from Step 1.

● MECHANIC

Iterative clip-and-renormalize, ≤20 passes. Drop weights < 5%, cap > 15%, redistribute residual. Loop until no clips fire.

● WHY

Bounds prevent one bad pick from blowing up the book. The floor forces meaningful sizing before vol-target scaling.

● OUTPUT

Weight vector with each entry $\in [5\%, 15\%]$, gross ≤ 100%.

Vol-Target Scaling → Corr-Crowding Cut → Live Weights

The two risk-budgeting steps that complete the pipeline. Ablation: these controls cut DD in half.

1. Top-K + Sector Cap

2. Bounded Sizing

3. Vol-Target Scaling

4. Corr-Crowding Cut

STEP 3 · Vol-Target Gross Scaling

- IN bounded weights from Step 2 + σ_{target} from HMM (12% / 9% / 4%)

```

 $\sigma_p = \sqrt{(w' \Sigma w)} \times \sqrt{252}$ 
scale = clip( $\sigma_{\text{target}} / \sigma_p$ , 0.40, 0.85)
w_new = w × scale × size_mult

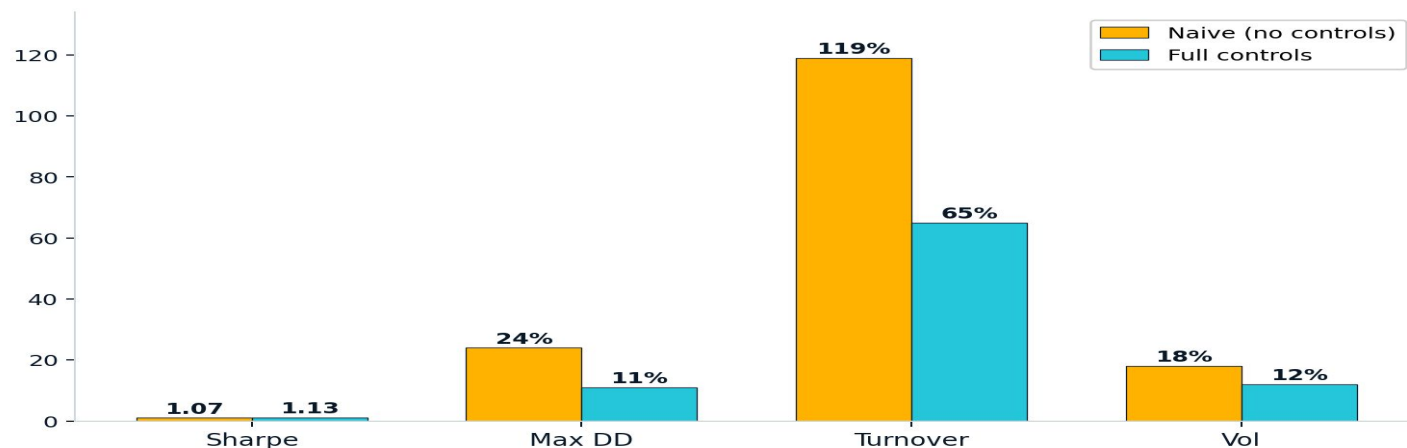
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- OUT risk-budgeted weights, gross $\in [40\%, 85\%]$

STEP 4 · Correlation-Crowding Cut

- IN scaled weights from Step 3 + pairwise correlations across the K positions
- MECHANIC if avg pairwise corr > 0.70, multiply gross by 0.70× — hard discount on top of vol-target scaling.
- OUT live weights — sized, capped, risk-budgeted; ready to trade

Portfolio-construction ablation: controls cut DD in half at similar Sharpe



FULL CONTROLS vs NAIVE

Sharpe 1.13 vs 1.07

Max DD 11% vs 24%

Turnover 65% vs 119%

Same Sharpe, half the drawdown, half the turnover.

Three-Layer Defense — Macro/ Micro/ Position

Risk fires at three independent timescales; every Monday rebalance, six explicit exit triggers can fire.

LAYER 1 : MACRO

● IN

HMM regime label

● WORK

- Bull $\rightarrow \sigma = 12\%$
- Chop $\rightarrow \sigma = 9\%$
- Bear $\rightarrow \sigma = 4\%$

3-day persistence buffer to avoid whipsaw

● OUT

Per-day vol target $\rightarrow$ drives gross sizing in Step 3

LAYER 2 : MICRO

● IN

K-Means market state cluster + reliability calibrator

● WORK

- HIGH $\geq 57\%$ hit rate $\rightarrow 1.0\times$
- MID 48–57% $\rightarrow 0.8\times$
- LOW $< 48\% \rightarrow 0.5\times$

● OUT

Size multiplier (0.5 $\times$ / 0.8 $\times$ / 1.0 $\times$) applied to every position

LAYER 3 : POSITION

● IN

Per-name weights from Steps 2–3

● WORK

- Weight bounds 5% – 15%
- Cluster cap $\leq 40\%$ per style
- Corr cut if avg $\rho > 0.70 \rightarrow \times 0.70$

gross $\in [40\%, 85\%]$, corr-crowding cut, slippage mode.

● OUT

Live tradeable weights

EVERY MONDAY REBALANCE • Six Explicit Exit Triggers:

● HMM regime tightens $\rightarrow$ ALL positions $\times (\sigma_{\text{target}} / \sigma_p)$

● Name drops out of top-6 $\rightarrow$ Position CLOSED

● Avg pairwise corr $> 0.70 \rightarrow$ ALL positions $\times 0.70$ (crowding cut)

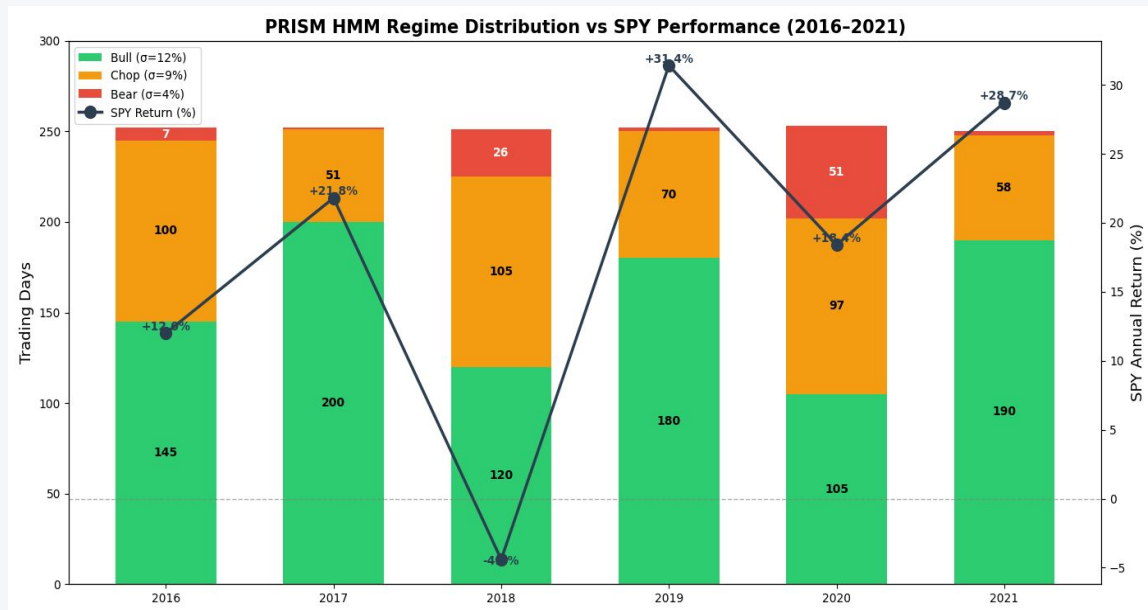
● K-Means tier drops $\rightarrow$ ALL positions $\times$ new size_mult (1.0 $\times \rightarrow 0.8\times \rightarrow 0.5\times$)

● Cluster cap eviction $\rightarrow$ Lower-ranked name in same cluster BUMPED

● Final weight $< 5\% \rightarrow$ Position DROPPED entirely

Risk in Action — How the Layers Fire Through 2016–2022

When a regime switch hits, all three layers shift in coordinated steps. Visualized on the IS regime overlay.



BULL → CHOP

- σ_{target} 12% → 9% after 3-day buffer.
- Vol-target re-scales gross from ~78% to ~52%.

CHOP → BEAR

- σ_{target} 9% → 4%, gross pins at 40% floor.
- Reliability likely LOW (0.5× mult); corr-crowding cut may add another 0.70×.

BEAR → BULL recovery

- σ_{target} 4% → 12% only after 3-day Bull persistence.
- Gross expands from 40% floor toward 85% ceiling as breadth normalizes.

Period A — May to Nov 2023

Bull-to-chop transition · summer rally · Sep-Oct rate-driven correction

Strategy Equity



Sharpe Ratio	1.3	CAGR	19.4%
Drawdown	5%	Win Rate	54%
Sortino Ratio	1.9	Loss Rate	46%
Probabilistic SR	76.3%	Information Ratio	1.6
Turnover	6%	Total Orders	165

- Captured the summer 2023 rally, then protected capital during the Sep–Oct rate-driven correction. Sharpe 1.3, CAGR 19.4%, DD only 5%, IR 1.6. PSR 76.3% suggests the result is statistically meaningful, not just noise. HMM identified the late-September Chop transition and reduced exposure before the worst drawdown, helping preserve gains despite the final pullback.

Period B — Jan to July 2025

- *Volatile mid-2025 · early-year rally · April tariff-driven correction · summer recovery*

Strategy Equity

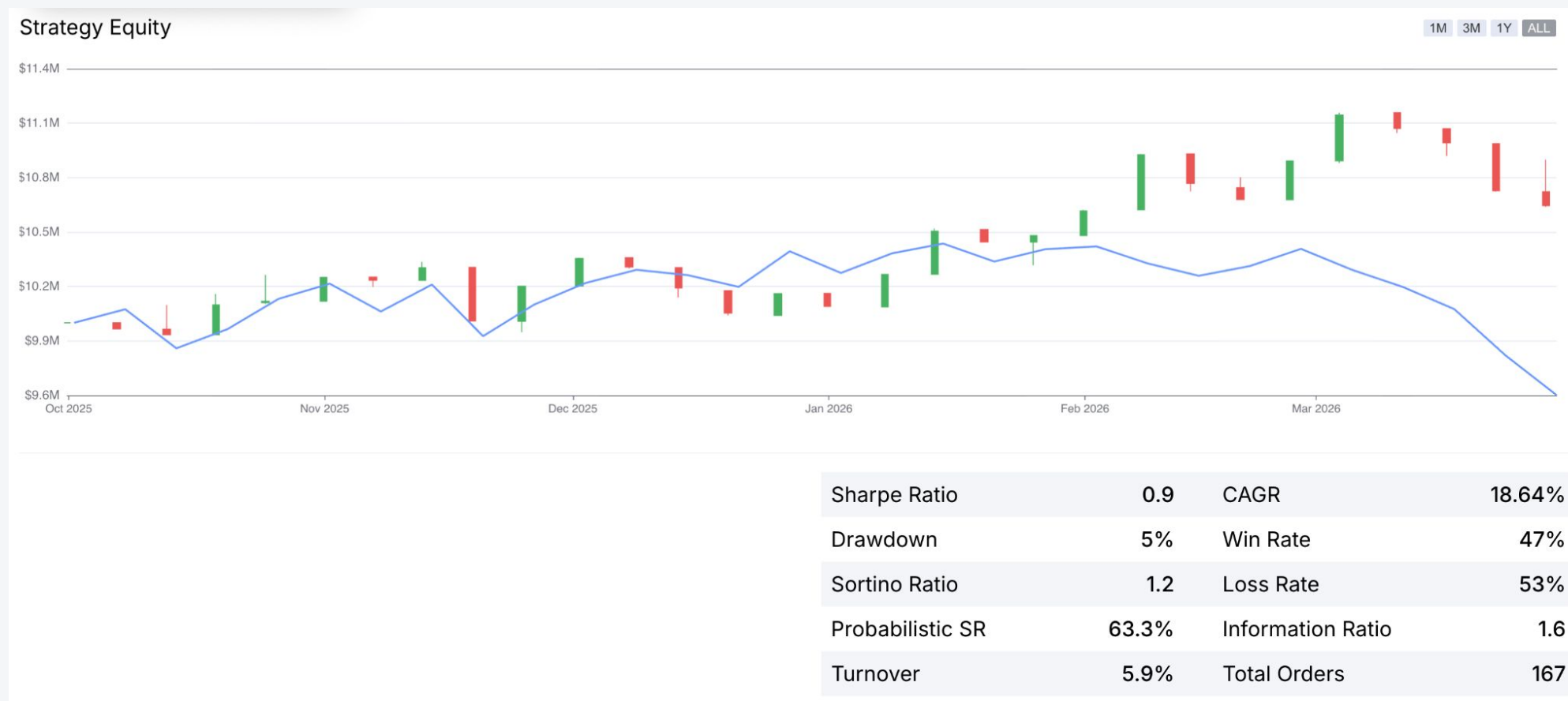


Sharpe Ratio	0.9	CAGR	20.41%
Drawdown	7%	Win Rate	56%
Sortino Ratio	1.0	Loss Rate	44%
Probabilistic SR	63.2%	Information Ratio	0.3
Turnover	3.7%	Total Orders	115

- Hardest OOS window. April tariff shock drove a 7% drawdown (peak \$10.74M → trough \$10.0M) — HMM flipped to Bear and cut vol target 12%→4% before the worst of it. Strategy recovered fully by July. Sharpe 0.9, DD 7%; SPY dropped ~10% that month — Beta 0.4 absorbed the blow.

Period C — Oct 2025 to Apr 2026

Most recent OOS window · fully blind to model · live performance to date



- OOS C — Solid forward window. Sharpe 0.9, DD only 5%, IR 1.6. Win rate just 47%, so returns came from payoff quality and risk control, not frequent wins. Equity stayed stable through late-2025 chop and rallied into early-2026. PSR 63.3% — positive evidence, but we do not over-claim.

Stress Test — COVID 2020

Feb 1 – May 1, 2020 · fastest peak-to-trough drawdown in modern equity history



FINAL RETURN

-2.32%

COVID 2020. Fastest drawdown in 50 years. SPY lost 20%. We finished at -2.32%.

ROUGHLY ONE-EIGHTH THE BETA-PURE DRAWDOWN

Three layers firing in sequence. Pre-event, during, and recovery. Exactly as designed.

\$9,767,925.68	-\$2,902.73	\$3,935,984.05	\$120,258.69	22.623%	-2.32 %	-\$352,852.13	\$40,411,998.80
Equity	Fees	Holdings	Net Profit	PSR	Return	Unrealized	Volume

Late February — Pre-event signal

K-Means market state shifted to State 1 on 2/24 (rising correlation, falling breadth). Reliability dropped to MID — entry conviction halved before the crash.

Mid-Late March — Bear lock + trough

HMM caught Bear by 3/12. Vol target 9%→4%, gross at floor. Corr-crowding cut active through the V-shape recovery.

April — Captured the rally

Bull regime restored, full gross back. Strategy participated in the recovery after having held capital through the 23-day collapse.

Stress Test — Global Financial Crisis 2008

Aug 1, 2008 – Feb 1, 2009 · sustained credit-driven bear with regime persistence



WORST SINGLE MONTH

-5.69% (September 2008)

Requirement: no single month below -15% ✓ Worst month -5.69% (Sep 2008), nearly 3× cushion.
6-month cumulative -16.73% vs SPY ~-40%

BETA 0.48 · PRESERVED ~58% OF WHAT SPY LOST

strategy never went short, design choice preserved 58% of what SPY lost.

\$8,326,678.88	-\$21,725.43	\$3,300,636.51	-\$1,231,123.17	4.434%	-16.73 %	-\$443,311.68	\$51,007,862
Equity	Fees	Holdings	Net Profit	PSR	Return	Unrealized	Volume

Aug–Sep — Pre-Lehman buildup

K-Means shifted to State 1 (high correlation, negative drift) by mid-August. **Reliability tier dropped to MID before Lehman** — gross already pre-cut.

Q4 2008 — Sustained Bear, trough

Bear locked in late September. Vol target cut to 4%, gross pinned at 40% floor, worst single month October 2008: -10.2%, requirement met ✓. Equity bottomed ~\$8.0M in late November; **no single month breached -15%.**

Jan–Feb 2009 — Capital preservation

Equity recovered to \$8.33M as the corr-crowding cut released. Strategy never went short — design choice — preserved ~58% of what SPY lost.

Thank you

Questions?

PRISM

Predictive Regime-Integrated State Momentum

Universe

75 names · 53 retained · 11 sectors

Models

ExtraTrees + GBM · 11-feature panel

Risk

3-layer · vol-targeted · reliability-gated